Maha Sajid

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Education

Virginia Tech Expected May 2028

Ph.D. Computer Science · CGPA: 4.00/4.00

- Research Focus: Multimodal Behavioral Modeling, Eye Tracking, Human Attention, Collaborative XR

Lahore University of Management Sciences (LUMS) Aug 2018 – May 2022

B.S. Computer Science · CGPA: 3.72/4.00

- Minor in Psychology · Graduated with Distinction (Top 10% students)

Research Experience

Graduate Research Assistant Aug 2023 – Present

Virginia Tech, Computer Science *Blacksburg, VA*

Advisors: Dr. Brendan David-John and Dr. Bo Ji

Predictive Modeling of Team Attention from Eye Tracking (ongoing):

- Built a **three-person VR environment** on Samsung Galaxy XR with asymmetric roles and interdependent decision-making tasks to investigate team attention allocation and coordination under varying task demands.
- Designed and am conducting a **repeated-measures human-subject study**, linking continuous eye-tracking and task data to periodic NASA-TLX and SART ratings to characterize attention, workload, and situational awareness.
- Developing **predictive ML models** by extracting and comparing individual- and team-level gaze features to infer attentional states, with planned cross-participant and cross-condition validation to assess generalizability.

Gaze and Voice Communication in Collaborative MR (under review, IEEE VR 2027):

- Led end-to-end research on human attention and SA in a collaborative **seek-and-shoot MR game**, building a networked HoloLens 2 system and conducting a **12-participant pilot** and a **48-participant (24-dyad) study**.
- Developed **multimodal temporal analysis pipelines** by synchronizing eye-tracking, speech, and gameplay data to derive gaze entropy, joint attention, speaker switches, and speech-to-gaze orientation latency for quantifying communication and coordination.
- Extended team situational-awareness assessment by deriving **information coverage and accurate alignment** from paired SAGAT responses to distinguish distributed task knowledge from accurately shared understanding.
- Evaluated three communication modalities using behavioral and awareness measures, finding that **Gaze+Voice improved task performance** over Voice-only and accurately shared awareness over Gaze-only.

Security Attacks in Remote Collaborative MR (published, IEEE VR 2025):

- Led a **mixed-methods behavioral study**, refining the experimental protocol with 16 pilot participants and evaluating 20 participants (10 dyads) to investigate how users perceive and respond to security attacks in MR.
- Developed **four real-time security attack scenarios**, including spatial occlusion and 3D click-redirection, in a networked MR environment to examine how interface disruptions affect user awareness and behavior.
- Analyzed first-person recordings, questionnaires, and interviews to identify gaps between perceived and actual user behavior, finding that **54% reported no attack** during spatial occlusion.
- Translated quantitative and qualitative findings into **design recommendations** for security indicators, interaction transparency, and user guidance to improve awareness of malicious interactions in MR.

Multimodal Privacy Indicators Under Situational Impairment (IEEE ISMAR Adjunct 2025):

- Collaborated on a mixed-methods study investigating how situational impairment affects the usability of privacy indicators, using four focus groups to develop five visual and multimodal concepts, followed by participant evaluation.
- Analyzed participant feedback to compare indicator effectiveness across contexts, finding that visual-only cues were less effective under situational impairment while multimodal indicators were preferred in privacy-sensitive scenarios.

Undergraduate Research Assistant Jan 2022 – Feb 2023

Lahore University of Management Sciences (LUMS), Computer Science *Lahore, Pakistan*

Advisors: Dr. Fareed Zaffar and Dr. Murtaza Taj

Cross-Verification of Revenue Using Remote Sensing Data:

- Developed and evaluated a **computer-vision pipeline** for satellite-image segmentation and land-use change detection, achieving **92% accuracy** in identifying constructed areas.

- Developed a computer-vision pipeline for satellite-image segmentation and land-use change detection, **achieving 92% accuracy** in identifying constructed areas for property verification.
- Built a **GIS-integrated decision-support platform** translating ML outputs into spatial and temporal construction comparisons across locations and years.
- Integrated remote-sensing outputs into tax-record cross-verification workflows to flag construction inconsistencies and inform field surveys; contributed to an **International Growth Centre technical report**.

Selected Publications

- **Sajid, M.**, Bukhari, S. I. M. S., Ji, B., & David-John, B. (2025). *"Just stop doing everything for now!": Understanding Security Attacks in Remote Collaborative Mixed Reality.* **IEEE VR 2025**. DOI: 10.1109/VR59515.2025.00085.
- Bukhari, S. I. M. S., **Sajid, M.**, Ji, B., & David-John, B. (2025). *Rethinking Privacy Indicators in Extended Reality: Multimodal Design for Situationally Impaired Bystanders.* **IEEE ISMAR Adjunct 2025**. DOI: 10.1109/ISMAR-Adjunct68609.2025.00059.

Industry Experience

Software Engineer Feb 2023 – Jul 2023
Senarios *Lahore, Pakistan*
- Delivered a cross-platform mobile data-sharing product from design through release, implementing real-time contact synchronization, decentralized functionality for a Google Play deployment.
- Partnered with **3+ clients** to translate Web3 data-integrity requirements into implementation plans and coordinate feature delivery across engineering and product stakeholders in an Agile environment.

Software Engineer Jun 2022 – Jan 2023
PakWheels *Lahore, Pakistan*
- Developed a **Python/XGBoost vehicle-pricing model** using marketplace data to provide data-driven price recommendations.
- Led payment-gateway and checkout integration for an automotive marketplace from implementation through production deployment.

Teaching Experience

Graduate Teaching Assistant Dec 2024 – Dec 2025
Virginia Tech
- Courses: CS 1054: Intro to Programming in Java; CS/ECE 5590: System and Software Security; CS 5024: Ethics and Professionalism in Computer Science.

Undergraduate Teaching Assistant Aug 2022 – May 2023
LUMS
- Courses: CS 360: Software Engineering; CS 5310: Computer Vision Fundamentals.

Research & Technical Skills

Programming & ML: Python, Pandas, NumPy, TensorFlow, XGBoost, classical and deep learning, predictive modeling, feature engineering, model training and evaluation, Git

Multimodal & Sequential Data Analysis: Eye tracking, time-aligned gaze/speech/task streams, gaze-object mapping, joint attention, gaze entropy, visual coverage, speech turn-taking, verbal coordination, behavioral telemetry

Experimental Methods & Statistics: Human-subject studies, repeated-measures designs, SAGAT, NASA-TLX, SART, mixed methods, ANOVA, Friedman and Wilcoxon tests, bootstrapping, interviews, thematic analysis

Computer Vision & XR Systems: Image segmentation, object detection, Unity, OpenXR, HoloLens 2, Samsung Galaxy XR, networked XR, C#, C++

Honors, Leadership & Service

- **Awards & Fellowships:** John Lee Pratt Fellowship (2026–2027) · Two Commonwealth Cyber Initiative Cyber Innovation Scholar Awards ($3,500, 2024–2026) · TAPIA Scholarship (2026).
- **Peer Review & Service:** Reviewer, IEEE VR (2026) and Human-Centric Intelligent Systems (2026) · Organizing Committee, iXR Ideathon (2026) · IEEE ISMAR Student Volunteer (2025).
- **Mentorship:** Mentored two Virginia Tech undergraduate researchers; Computing Research Association (CRA) UR2PhD Certified Graduate Mentor (2026).
- **Leadership & Honors:** Graduate Council Officer, Virginia Tech CS (2026–2027) · General Secretary, LUMS Data Science Society (2021–2022) · Dean's Honor List, LUMS (2019–2022).